

Library Management System - Lab Work

1. Create a database called `LibraryDB`:

```
CREATE DATABASE LibraryDB;
```

2. Create a table titled `Members`:

```
CREATE TABLE Members (  
  member_id INT PRIMARY KEY AUTO_INCREMENT NOT NULL,  
  name VARCHAR(100) NOT NULL,  
  email VARCHAR(100) UNIQUE NOT NULL,  
  phone VARCHAR(20) NOT NULL,  
  address TEXT NOT NULL  
);
```

3. Create a table titled `Books`:

```
CREATE TABLE Books (  
  book_id INT PRIMARY KEY AUTO_INCREMENT NOT NULL,  
  title VARCHAR(150) NOT NULL,  
  author VARCHAR(100) NOT NULL,  
  genre VARCHAR(50) NOT NULL,  
  available_copies INT NOT NULL  
);
```

4. Create a table titled `BorrowRecords`:

```
CREATE TABLE BorrowRecords (  
  record_id INT PRIMARY KEY AUTO_INCREMENT NOT NULL,  
  member_id INT NOT NULL,  
  borrow_date DATE NOT NULL,  
  return_date DATE,  
  FOREIGN KEY (member_id) REFERENCES Members(member_id)  
);
```

5. Create a table titled `BorrowedBooks`:

```
CREATE TABLE BorrowedBooks (  
  entry_id INT PRIMARY KEY AUTO_INCREMENT NOT NULL,  
  record_id INT NOT NULL,
```

```
book_id INT NOT NULL,  
quantity INT NOT NULL,  
FOREIGN KEY (record_id) REFERENCES BorrowRecords(record_id),  
FOREIGN KEY (book_id) REFERENCES Books(book_id)  
);
```

Queries

Query 1 (Easy): Show all books under the "Science Fiction" genre.

```
SELECT * FROM Books WHERE genre = 'Science Fiction';
```

Query 2 (Easy): List all borrow records before March 1, 2025.

```
SELECT * FROM BorrowRecords WHERE borrow_date < '2025-03-01';
```

Query 3 (Medium): Show name and email of members who have borrowed any book.

```
SELECT DISTINCT M.name, M.email  
FROM Members M  
JOIN BorrowRecords B ON M.member_id = B.member_id;
```

Query 4 (Medium): Find the titles and quantities of books borrowed in record ID = 2.

```
SELECT B.title, BB.quantity  
FROM BorrowedBooks BB  
JOIN Books B ON BB.book_id = B.book_id  
WHERE BB.record_id = 2;
```

Query 5 (Medium-Hard): Find the total number of books borrowed by each member.

```
SELECT M.name, SUM(BB.quantity) AS total_books_borrowed  
FROM Members M  
JOIN BorrowRecords BR ON M.member_id = BR.member_id  
JOIN BorrowedBooks BB ON BR.record_id = BB.record_id  
GROUP BY M.member_id;
```

Query 6 (Medium-Hard): Find the most recent borrowing with the member name.

```
SELECT M.name, BR.borrow_date
FROM BorrowRecords BR
JOIN Members M ON BR.member_id = M.member_id
ORDER BY BR.borrow_date DESC
LIMIT 1;
```

Query 7 (Medium-Hard): List books that have been borrowed more than once.

```
SELECT B.title, COUNT(*) AS times_borrowed
FROM BorrowedBooks BB
JOIN Books B ON BB.book_id = B.book_id
GROUP BY B.book_id
HAVING COUNT(*) > 1;
```

Sample Data Insertions

Members Table:

```
INSERT INTO Members (name, email, phone, address) VALUES
('Tania Ahmed', 'tania@mail.com', '+8801712000001', 'Dhaka,
Bangladesh'),
('Fahim Islam', 'fahim@outlook.com', '+8801712000002', 'Sylhet,
Bangladesh'),
('Nusrat Jahan', 'nusrat@gmail.com', '+8801712000003',
'Chittagong, Bangladesh'),
('Asif Rahman', 'asif@yahoo.com', '+8801712000004', 'Barisal,
Bangladesh'),
('Karim Hossain', 'karim@mail.com', '+8801712000005', 'Rajshahi,
Bangladesh');
```

Books Table:

```
INSERT INTO Books (title, author, genre, available_copies) VALUES
('The Time Machine', 'H.G. Wells', 'Science Fiction', 5),
('Sapiens', 'Yuval Noah Harari', 'History', 7),
('To Kill a Mockingbird', 'Harper Lee', 'Fiction', 3),
('1984', 'George Orwell', 'Dystopian', 4),
('The Alchemist', 'Paulo Coelho', 'Adventure', 6);
```

BorrowRecords Table:

```
INSERT INTO BorrowRecords (member_id, borrow_date,  
return_date) VALUES  
(1, '2025-02-20', '2025-03-01'),  
(2, '2025-02-22', NULL),  
(3, '2025-02-25', NULL),  
(4, '2025-02-28', '2025-03-05'),  
(5, '2025-03-01', NULL);
```

BorrowedBooks Table:

```
INSERT INTO BorrowedBooks (record_id, book_id, quantity) VALUES  
(1, 1, 1),  
(2, 2, 1),  
(2, 3, 1),  
(3, 1, 1),  
(3, 4, 1),  
(5, 5, 1);
```
